Tribhuvan University

Faculty of Humanities and Social Science

A Final Year Internship Report On "FRONTEND DEVELOPMENT" of Financial Cooperative Registration System (FCRS) for National Cooperative Regulatory Authority (NCRA) at DEBUGSOFT PVT. LTD.

Under the supervision of

Abiral K.C.

In partial fulfillment of the requirement for a Bachelor in Computer Application (BCA), Seventh Semester

Submitted By:

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Submitted To:

Tribhuvan University

Faculty of Humanities and Social Science

Department of Bachelor in Computer Application

Kirtipur, Kathmandu

April, 2026

DECLARATION

The internship report entitled "**FRONTEND DEVELOPMENT" of Financial Cooperative Registration System (FCRS) for National Cooperative Regulatory Authority (NCRA) at DEBUGSOFT PVT. LTD** is being submitted to the Department of Computer Application, Dillibazar, Kathmandu, Nepal, under the supervision of **Mr. Abiral K.C**. This project is original and has not been submitted earlier in part or full in this or any other form to any university or institute, here or elsewhere, for the award of any degree.

Simran Pun (6-2-354-25-2021)

CORPORATE CERTIFICATE

Regd. No.: 114747/070/071

Debugsoft P. Ltd.

Kupondole, Lalitpur, Nepal

31st March, 2026

To Whom It May Concern,

This is to certify that **Simran Pun** has successfully completed internship as a **Frontend Developer** in React at DEBUGSOFT Pvt. Ltd., from **October 25, 2025 to February 25, 2026.**

During the internship, **Simran Pun** contributed significantly to the Web Application Development project by developing a Financial Cooperative Registration System (FCRS) for National Cooperative Regulatory Authority (NCRA), integrating API, designing UX/UI for the application, and testing applications. Her work enhanced our system's usability capabilities.

We commend her technical expertise, professionalism, and effective communication skills. It has been a pleasure having her on our team, and we are confident that she will continue to excel in her future endeavors.

Sincerely,

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MENTOR'S RECOMMENDATION

I hereby recommend that this report prepared under my mentorship by **Simran Pun** entitled "**FRONTEND DEVELOPMENT" of Financial Cooperative Registration System (FCRS) for National Cooperative Regulatory Authority (NCRA) at DEBUGSOFT PVT. LTD** in partial fulfillment of the requirements for the degree of Bachelor in Computer Application (BCA) of Tribhuvan University is processed for the evaluation.

.....

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SUPERVISOR RECOMMENDATION

This is to recommend that **Simran Pun** carried out an internship report entitled "**FRONTEND DEVELOPMENT" of Financial Cooperative Registration System (FCRS) for National Cooperative Regulatory Authority (NCRA) at DEBUGSOFT PVT. LTD** for the fulfillment of the seventh semester under my supervision. To my knowledge, this work has not been submitted for any other degree. He has fulfilled all the requirements laid down by the Trinity International College Department of Bachelor in Computer Application, Dillibazar, Kathmandu.

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LETTER OF APPROVAL

Date: April 2026

On the recommendation of **Abiral K.C.**, this Project Report submitted by **Simran Pun** entitled **"FRONTEND DEVELOPMENT" of Financial Cooperative Registration System (FCRS) for National Cooperative Regulatory Authority (NCRA) at DEBUGSOFT PVT. LTD** in partial fulfillment of the requirement for the award of the bachelor's degree in Computer Application is a bona fide record of the work carried out under my guidance and supervision at Trinity International College, Dillibazar, Kathmandu.

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ACKNOWLEDGEMENT

I am sincerely grateful to DEBUGSOFT Pvt. Ltd. for providing me with the opportunity to serve as a Frontend Intern from October 25, 2025 – February 25, 2026. This internship has been an invaluable experience, allowing me to gain practical knowledge and skills in Frontend development with React. I would like to extend my heartfelt thanks to the entire team at DEBUGSOFT Pvt. Ltd., particularly those at the Kupondole, Lalitpur office, for their support and guidance throughout my internship. The welcoming and collaborative environment significantly contributed to my learning and professional development. I am also immensely grateful to **Mr. Jenish Sambahamphe** my mentor, for his dedicated guidance during my internship. His expertise and mentorship have been instrumental in shaping my understanding of frontend technologies and processes. I am especially thankful to **Mr. Abiral K.C.** my supervisor, for his continuous support and valuable feedback, which were essential to my growth and understanding of frontend development workflows. Finally, I would like to express my gratitude to the Executive Director of DEBUGSOFT Pvt. Ltd. **Mr. Suresh Basnet** for effective communication and assistance during the onboarding process and throughout my tenure.

Thanking You,
Simran Pun (TU Roll No.: 6-2-354-25-2021)

ABSTRACT

This report details The Author's work on the Financial Cooperative Registration System (FCRS) for the National Cooperative Regulatory Authority (NCRA), a centralized digital platform designed to register, manage, and maintain cooperative data to enhance governance, transparency, and operational efficiency. The Author's primary responsibilities involved designing responsive user interfaces using React.js and Material UI, managing API requests with Axios, and developing dynamic forms, tables, and statistical reports. Contributions included Cooperative Registration, Infrastructure Management, Quality Assurance & Accreditation (QAA) systems, financial management components (Loan, Saving, Share), and advanced features such as print functionality, dynamic certificate templates, and drag-and-drop file uploads. These improvements enabled cooperative registrars, auditors, and managers to handle registration, compliance tracking, and financial data more effectively. The internship provided valuable practical experience in API integration, data visualization, dynamic form handling, responsive design, problem-solving, and collaborative development. Overall, this project highlights the importance of scalable, user-friendly systems for cooperative data management.

Keywords: *Cooperative Registration System, Cooperative Data Management, React.js, Frontend Development, API Integration, Financial Cooperative Registration*

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LIST OF ABBREVIATION

API:	Application Programming Interface
CMIS:	Cooperative Management Information System
CSS:	Cascading Style Sheets
FCRS:	Financial Cooperative Registration System
GIT:	Global Information Tracker (version control system)
ICT:	Information and Communication Technology
IoT:	Internet of Things
IT:	Information Technology
MUI:	Material-UI (a React UI library)
NCDB:	National Cooperative Development Board
NCRA:	National Cooperative Regulatory Authority
OTP:	One-Time Password
PDF:	Portable Document Format
QAA:	Quality Assurance & Accreditation
SDLC:	Software Development Life Cycle
UI/UX:	User Interface / User Experience

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CHAPTER 1: INTRODUCTION

1.1 Introduction

This report details The Author's 10-week internship focused on frontend development for the Financial Cooperative Registration System – National Cooperative Regulatory Authority (FCRS - NCRA) project, a National Cooperative Development Board (NCDB) initiative to centralize Nepal's cooperative sector data (N. C. R. Authority, 2074) . The Author contributed to designing and refining user interface components using React.js and Material UI, developing forms and tables for Cooperative Infrastructure Management, Quality Assurance & Accreditation (QAA) systems, and financial modules. Specific implementations included print functionality for official documents, dynamic certificate templates, drag-and-drop file uploads, and Axios-based API integration for seamless data exchange. The Author also generated statistical reports consolidating cooperative data, member demographics, and financial activities in tabular format. Additionally, role-based access control views were implemented to ensure that cooperative administrators and regulatory staff saw only relevant data and actions. Error handling and form validation were incorporated to improve data integrity and reduce submission errors during cooperative registration and compliance reporting. The frontend was optimized for responsive design, ensuring accessibility across desktops, tablets, and mobile devices used in semi-urban areas. The Author also collaborated with backend developers to debug API inconsistencies and participated in weekly sprint reviews to align deliverables with project milestones. The internship provided practical experience in modern frontend development, API integration, and responsive design, ultimately contributing to a centralized digital system that improves transparency, accountability, and operational efficiency for Nepal’s cooperative regulators and institutions.

1.2 Problem Statements

The cooperative sector is a critical pillar of Nepal's economy, providing financial inclusion to millions in rural and underserved communities. However, its management and regulation remain hindered by fragmented, paper-based processes. Currently, cooperative registration, financial reporting, and compliance monitoring are largely manual, leading to several critical issues: data fragmentation across thousands of cooperatives causing inconsistent record-keeping; absence of a centralized digital platform delaying registrations, complicating audits, and hindering regulatory monitoring; and manual data handling increasing risks of errors, duplication, and loss of financial records. Regulatory authorities also face challenges in generating timely, accurate reports on cooperative performance and member demographics, which limits effective policy formulation

and identification of at-risk cooperatives. For administrators, the lack of digital tools for managing member records and financial transactions results in operational inefficiencies. Additionally, ensuring transparency and accountability remains a pressing concern, as without a unified system to track ledgers and transactions, both cooperatives and regulators struggle to detect irregularities and prevent fraud. The FCRS - NCRA project seeks to address these challenges by developing a comprehensive digital system that centralizes cooperative data and automates key processes. This internship focuses on contributing to the frontend development of this system, ensuring an intuitive and efficient user interface for both cooperative administrators and regulatory authorities (Rising Nepal Daily, 2025) .

1.3 Objectives

The main objectives of this project are:

- To **establish a centralized digital platform** to eliminate manual, paper-based processes and ensure accurate, real-time data management for all cooperative institutions across Nepal (N. C. R. Authority, 2074).
- To **enhance transparency and accountability** in cooperative operations and financial activities by digitizing records and providing an unified system for tracking registrations, transactions, and compliance data.
- To **support regulatory oversight** by enabling the National Cooperative Regulatory Authority (NCRA) to monitor, supervise, and enforce compliance through an efficient online registration and approval process.
- To **provide data-driven insights and reporting tools** to facilitate better policy formulation, decision-making, and risk management within the cooperative sector.
- To **improve accessibility and scalability** by ensuring authorized stakeholders can efficiently access cooperative data while the system accommodates growing volumes of information reliably.

1.4 Scope and Limitation

1.4.1 Scope

The scope of the FCRS - NCRA project is to develop a comprehensive digital platform for the registration, management, and monitoring of cooperative institutions, ensuring improved governance, transparency, and regulatory compliance. From a frontend perspective, the scope includes designing responsive, user-friendly interfaces tailored to cooperative administrators, regulatory authorities, and other stakeholders. This involves developing dynamic stepper forms,

tables, and dashboards for structured data entry, management, and visualization of cooperative information. The system supports key functionalities such as cooperative registration, compliance tracking, and report generation, along with usability features including document generation, print functionality, file upload capabilities, and interactive interfaces. Overall, the project scope is centered on creating a scalable, accessible, and efficient platform that streamlines cooperative data management and supports effective decision-making and regulatory oversight within Nepal's cooperative sector.

1.4.2 Limitations

Potential performance constraints may occur during peak usage periods with multiple simultaneous users, affecting response times. Data synchronization issues could arise during high-volume data entry or external system integration, requiring careful optimization. Browser compatibility concerns may impact consistent rendering of advanced features on older browser versions. Security considerations related to financial data handling and user authentication demand rigorous implementation of industry-standard measures. Finally, the 10-week internship duration restricts the depth of feature implementation and comprehensive testing across all planned modules.

1.5 Report Organization

Report organization is an integral part of report formation. It gives the overall pattern of the report, which contains summary information of the comprehensive document. This document is categorized into several chapters and divided into sub-chapters, including all project details.

Chapter 1 – Introduction

Includes a short introduction of the system, problem statement, objectives, scope, and limitations of the FCRS - NCRA project.

Chapter 2 – Organization & Literature Review

Introduces DEBUGSOFT Pvt. Ltd., its hierarchy, and reviews literature related to cooperative management systems, frontend technologies, and regulatory frameworks.

Chapter 3 – Background Study

Presents the theoretical foundation and context for the project through a review of cooperative registration systems and related literature.

Chapter 4 – Intern Roles & Technical Activities

Lists the intern's roles and responsibilities, technical activities performed, tasks undertaken, and detailed descriptions of modules developed, including Cooperative Infrastructure Management, Financial Management Components, and Reporting Systems.

Chapter 5 – Conclusion & Future Enhancements

Covers learning outcomes, project conclusions, and potential future enhancements such as mobile application development, advanced analytics, and integration with other government systems.

CHAPTER 2: INTRODUCTION TO ORGANIZATION

2.1 Organization Details

DEBUGSOFT Pvt. Ltd. is a turnkey software and business solution provider established in 2013 with a clear vision: to steer customers through the next generation of business innovation powered by advanced technology. Co-founded by Mr. Suresh Basnet, who currently serves as the CEO, DEBUGSOFT has rapidly emerged as a leader in delivering state-of-the-art business automation, software development, and enterprise software solutions. Located in Sharada Marg Sankhamul (opposite of Kathmandu online cake shop), the company is dedicated to developing high-quality software that meets modern standards and focuses on creating user-friendly solutions.

Figure 2.1: DEBUGSOFT Pvt. Ltd.

Over the years, DEBUGSOFT has diversified its expertise to cover a broad range of services, including ICT consultation, telecommunication, networking, and IoT solutions. Currently, company is more focused on delivering IT services through web and app development.

The company is committed to supporting various business industries by providing comprehensive automation and solution products that address contemporary challenges. With a strong emphasis on innovation, DEBUGSOFT leverages the latest technologies to enhance business processes and drive digital transformation for its clients. The organizational structure at DEBUGSOFT is designed to foster collaboration and efficiency. With clear reporting lines and specialized teams, the company ensures that each project is managed with precision and meets rigorous quality standards. This structure not only supports robust project management but also enables the rapid development of tailored solutions for diverse business needs. As a result, DEBUGSOFT stands out as a dynamic and responsive organization dedicated to empowering businesses through cutting-edge technological solutions.

Table 2.1: Organization Details

Official Name	DEBUGSOFT Pvt Ltd.
Type of Business	IT
Address	Sharada Marg Sankhamul (opposite of Kathmandu online cake shop)
Email	dibugsoft@gmail.com
Key Service Areas	Web Development, App Development, IT Solutions
Staff Size	10 Full Time Employees and 4 Part Time Employees
Official Name	DEBUGSOFT Pvt Ltd.

2.2 Organizational Hierarchy

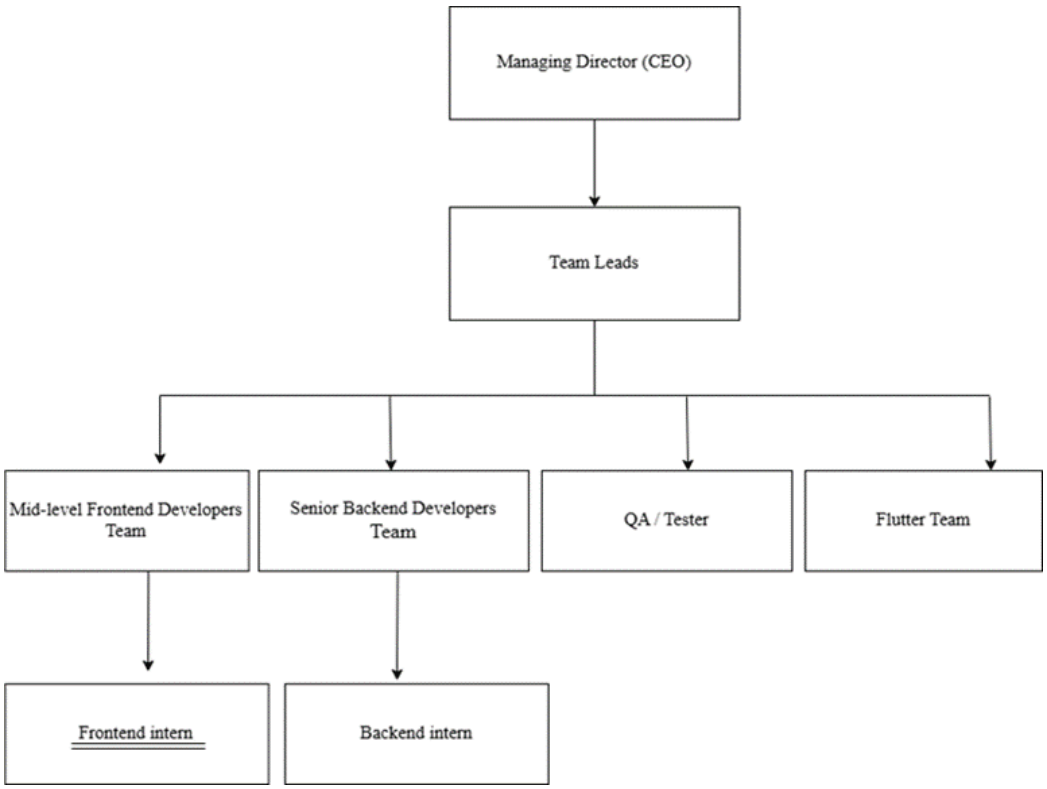


Figure 2.2: Organizational Hierarchy

2.3 Working Domains of Organization

DEBUGSOFT Pvt. Ltd. operates across a focused range of technology domains, primarily catering to government and private sector clients. The company's core working domains include:

- **Government Web Application Development** – Designing and developing secure, scalable web platforms for government agencies, including cooperative registration systems, regulatory portals, and public service automation tools.
- **Private Sector Software Solutions** – Building enterprise-grade applications for private businesses, including automation tools and management systems.
- **Hosting and Domain Solutions** – Providing reliable web hosting, domain registration, and server management services to ensure application availability and performance.
- **IT Consultation** – Offering expert guidance on technology strategy, system architecture, digital transformation, and project implementation for both government and private organizations.
- **Mobile Application Development** – Developing cross-platform and native mobile applications to extend digital services to users on smartphones and tablets.

While the company has traditional expertise in business automation, telecommunication, networking, and IoT, its current focus is predominantly on IT solutions, with primary emphasis on software development for government and private sector clients.

2.4 Description of Intern Department

The internship period is three months at DEBUGSOFT Pvt. Ltd. The internship details in the organization are summarized in the following table.

Table 2.2: Internship Details

Internship Title	Frontend Developer Intern
Organization	Private Limited
Address	Sharada Marg Sankhamul (opposite of Kathmandu online cake shop)
Mentor	Mr. Jenish Sambahamphe
Time period	24th October to 24th Februrary
Hours per day	8 hours (10 AM – 6PM)
Office days	6 days (Sun - Fri)

CHAPTER 3: BACKGROUND STUDY AND LITERATURE REVIEW

3.1 Background Study

This Digital transformation in the cooperative sector significantly enhances regulatory compliance, financial transparency, and operational efficiency. The proposed cooperative management system is an intelligent digital platform using modern web technologies to improve regulatory oversight and cooperative operations. The system utilizes React.js for responsive user interfaces, Material UI for consistent user experience, and Axios for reliable API communication between frontend and backend services. A component-based architecture allows administrators to manage member records, fee structures, and financial ledgers efficiently, with modules for infrastructure management, and quality assurance accreditation. Dynamic stepper forms, tables, and printable reports facilitate streamlined data entry, visualization, and documentation. Advanced features include dynamic certificate generation, print functionality for official records, drag-and-drop file uploads for supporting documents, and statistical reporting modules that consolidate multi-cooperative data for policy formulation and risk assessment.

The system also incorporates role-based access control, ensuring that cooperative administrators, regulatory officers, and superusers have appropriate permissions tailored to their responsibilities. Real-time form validation reduces data entry errors during cooperative registration and financial reporting, while responsive design guarantees accessibility across desktops, tablets, and mobile devices used in remote areas. Error handling mechanisms provide user-friendly feedback when API requests fail, allowing users to retry or correct inputs without losing progress. The platform supports pagination and search functionality within data tables, enabling efficient navigation through large volumes of cooperative records. Additionally, session management and secure authentication protocols protect sensitive financial and member information from unauthorized access. According to the National Cooperative Development Board (N. C. R. Authority, 2074; Rising Nepal Daily, 2025), digitization of cooperative governance in Nepal is essential for promoting transparency, accountability, and sustainable growth. Platforms like FCRS - NCRA address challenges related to data fragmentation, manual record-keeping, and limited regulatory oversight, ensuring cooperatives operate within established guidelines and contribute meaningfully to Nepal's economic development.

3.2 Literature Review

Existing literature on cooperative management systems in Nepal highlights a strong need for digital transformation to address persistent inefficiencies, including data fragmentation, inconsistencies, and limited regulatory oversight. Research emphasizes that Cooperative Management Information Systems (CMIS) should prioritize regulatory compliance through real-time audit trails and automated reporting mechanisms, aligning closely with FCRS - NCRA objectives. Studies advocate for modern frontend technologies, particularly React.js, in governance systems, highlighting component-based architectures for building responsive, scalable interfaces in environments with limited technical resources. The integration of tools such as Material UI and Axios is identified as effective for managing complex data interactions and ensuring smooth frontend-backend communication. Component-based development has been shown to significantly reduce development time while improving data accuracy in large-scale systems handling extensive cooperative data. Additional research highlights workflow efficiency improvements through modern interface features such as simplified document handling and automated generation processes, which contribute to consistency and reduced processing time.

Further studies reinforce these findings. Centralized digital platforms have been shown to reduce redundancy and improve data integrity across cooperative networks, directly addressing fragmentation problems in Nepal's cooperative sector. Role-based access control is emphasized as critical in cooperative systems, ensuring that sensitive financial and member data is accessible only to authorized personnel. Drag-and-drop file uploads and dynamic certificate generation are identified as key usability features that reduce administrative burden and accelerate registration workflows. Statistical reporting modules with real-time data aggregation enable regulators to identify non-compliant cooperatives proactively rather than reactively. The National Cooperative Development Board (NCDB) recognizes FCRS - NCRA as a key national initiative for digitizing the cooperative sector, serving as a model for technology-driven solutions that enhance regulatory oversight and data management. Mobile-responsive designs are also recommended to ensure accessibility for cooperative staff operating from rural and semi-urban areas with limited infrastructure. Overall, the reviewed literature supports the adoption of modern frontend technologies and centralized digital systems in cooperative governance, validating the practical implementation of responsive interfaces, efficient data handling, and user-centric design within the FCRS - NCRA platform (D. P. Ltd., 2082; N. C. R. Authority, 2074) .

CHAPTER 4: INTERNSHIP ACTIVITIES

4.1 Roles and Responsibilities

As a frontend developer intern for the FCRS-NCRA project, the roles and responsibilities are as follows:

- Utilizing a search-ask-learn approach when acquiring knowledge about various frontend technologies essential for the project's development, including React.js, Material UI, and Axios.
- Promptly communicating project-related issues such as UI/UX inconsistencies, responsive design challenges, and component integration problems to the project supervisor or mentor.
- Ensuring timely completion of assigned frontend development tasks while adhering to the project schedule defined by the supervisor.
- Engaging in comprehensive study of modern frontend development principles, including component-based architectures, state management, and API integration, while exploring React ecosystem tools under mentor guidance.
- Assuming responsibility for the development and upkeep of the frontend interface, employing React.js, Material UI, and Axios to create responsive, user-friendly components for cooperative administrators and regulators.
- Fostering collaboration with backend teams and designers, contributing to efficient project development through active participation in cross-functional discussions and alignment on API contracts and UI requirements.

Initial Internship Phases

In the first phase, tasks involved getting familiar with the company environment, frontend developer workflow, and intern responsibilities at DEBUGSOFT Pvt. Ltd., including exploring React development tools, Vite setup, and Material UI documentation. The second week focused on discussions about the FCRS-NCRA technology stack, React component patterns, Material UI theming strategies, and setting up the React development environment with API integration using Axios.

4.2 Weekly Log

Table 4.1: Weekly Log

Weeks	Dates	Activities
Week 1	25 October – 31 October	• Understood project requirements and system workflow • Set up React project with TypeScript • Configured folder structure and project architecture • Installed dependencies (MUI, Tanstack Router, Axios) • Initialized Git repository
Week 2	1 November – 7 November	• Designed basic UI layout (Header, Sidebar, Footer) • Created reusable components (Button, Input, Card) • Implemented responsive layout using MUI Grid • Applied consistent color themes and typography
Week 3	8 November – 14 November	• Developed Login and Registration UI • Implemented form validation • Integrated authentication APIs • Managed user session and token storage
Week 4	15 November – 21 November	• Configured protected routes • Implemented role-based navigation • Developed dynamic sidebar menu • Improved navigation flow and UX
Week 5	22 November – 28 November	• Built admin dashboard UI • Displayed summary cards and statistics • Integrated API for dashboard data • Designed charts and visual components

Week 6	29 November – 5 December	• Developed cooperative registration form • Implemented multi-step form (Stepper using MUI) • Managed complex form state using React hooks • Ensured proper validation across steps
Week 7	6 December – 12 December	• Connected registration forms with backend APIs • Handled API responses and error states • Improved data handling and validation
Week 8	13 December – 19 December	• Implemented reusable hooks and utilities • Optimized component rendering • Refactored code for better maintainability
Week 9	20 December – 26 December	• Implemented file upload functionality with SSL heartbeat • Added preview and validation for documents • Integrated file APIs with real-time upload status • Ensured proper error handling
Week 10	27 December – 2 January	• Tested application across different devices • Fixed UI and functional bugs • Improved responsiveness • Debugged API integration issues
Week 11	3 January – 9 January	• Implemented lazy loading for components • Optimized API calls • Improved overall app performance • Enhanced UI/UX consistency
Week 12	10 January – 16 January	• Developed admin verification with approve/unapprove API integration and feedback • Created reusable components to fetch cooperative data for admin approval

		• Built dynamic tables: Registered List, Verified List, Approved List, Reports Table, Summary Reports • Implemented Forgot Password and Reset Password with full API integration • Implemented Change Password with API integration
Week 13	17 January – 24 January	• Final code cleanup and refactoring • Reviewed all modules and features • Assisted in deployment setup

4.3 Description of the project involved during Intern

During the internship at Debugsoft Pvt. Ltd., I worked on FCRS-NCRA (Financial Cooperative Registration System - National Cooperative Regulatory Authority), a transformative platform designed to centralize and digitize Nepal's cooperative sector data management. Developed under the National Cooperative Development Board (NCDB), FCRS-NCRA addresses critical challenges in cooperative governance by providing a comprehensive web-based system accessible at https://fcrs.ncra.gov.np (N. C. R. Authority, 2074) .

The project comprises multiple modules tailored for cooperative administrators, financial managers, and regulatory authorities, including:

- Cooperative Data Management
- Financial Management
- Quality Assurance Accreditation (QAA) systems
- Dynamic Reporting dashboards
- Document Management with dynamic certificates and print functionality

My role as Frontend Developer Intern focused on creating responsive, intuitive user interfaces using React.js and Material UI, ensuring seamless interaction across devices for cooperative stakeholders. Key contributions included developing dynamic forms and tables, integrating Axios for robust API communication, and implementing advanced features like drag-and-drop file uploads and printable certificate templates.

4.4 Tasks/ Activities Performed

During the author's internship as a Frontend Developer Intern at DebugSoft Pvt. Ltd., I was responsible for a range of tasks and activities that contributed to the development of the Financial Cooperative Registration System - National Cooperative Regulatory Authority (FCRS-NCRA). Below is a detailed breakdown of the tasks and activities performed:

i. Environment Setup and Skill Refreshment:

- Installed and configured essential tools: Node.js, Vite, and Git.

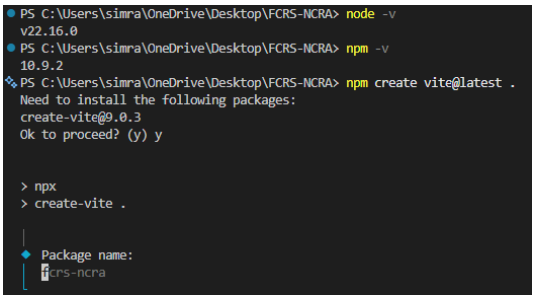


Figure 4.1: Installed and configured essential tools

- Set up a new React project with TypeScript for initial practice.

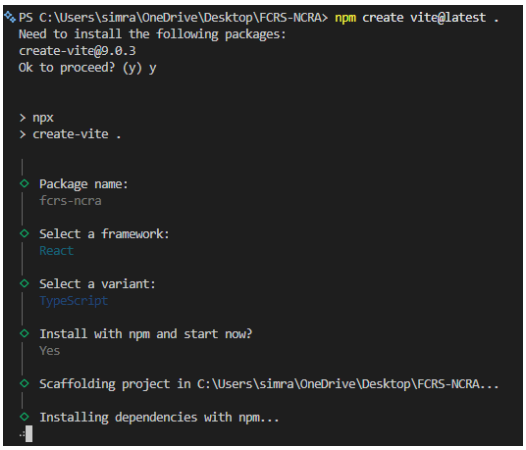


Figure 3.2: Set up a new React project

- Configured folder structure and project architecture.
- Installed dependencies: MUI, Tanstack Router, Axios.
- Initialized Git repository for version control.
- Revised fundamental JavaScript/TypeScript concepts (ES6+ features like arrow functions, async/await, promises).
- Refreshed core React principles (components, props, state management with useState, Tanstack Router).
- Simulated API calls using a JSON Server for initial development testing.

```
const fetchUser = async () => {
  if (!loggedInUserId || !token) return;

  try {
    const res = await axios.get(
      `${BASE_ENDPOINT}/User/${loggedInUserId}`,
      {
        headers: { Authorization: `Bearer ${token}` },
      }
    );

    const sig = res.data?.signatureUpload;
    setExistingSignature(buildFileUrl(sig));
  } catch (err) {
    console.error("Failed to fetch user", err);
  }
};
```

Figure 4.3: API calls using a JSON Server

ii. UI Layout and Reusable Components:

- Designed basic UI layout including Header, Sidebar, and Footer.

Figure 4.4: Header

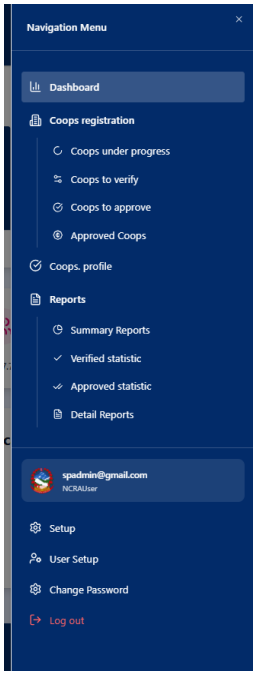


Figure 4.5: Sidebar

Figure 4.6: Footer

- Created reusable components (Button, Input, Card) for consistent design.

- Implemented responsive layout using MUI Grid system.
- Applied consistent color themes and typography aligned with NCRA branding.

iii. Authentication and Navigation:

- Developed Login and Registration UI with form validation.

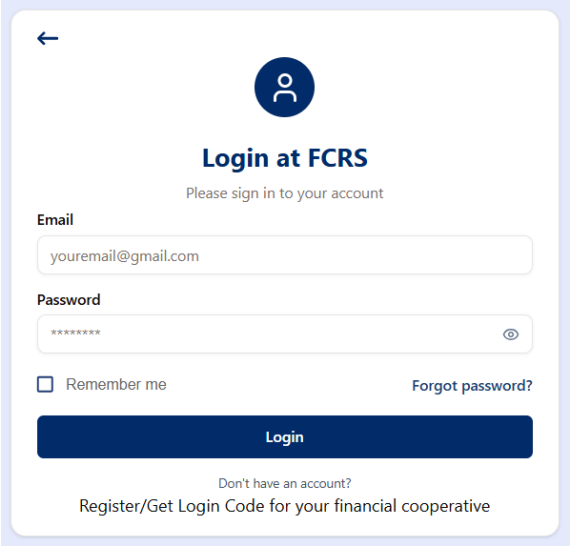


Figure 4.7: Login

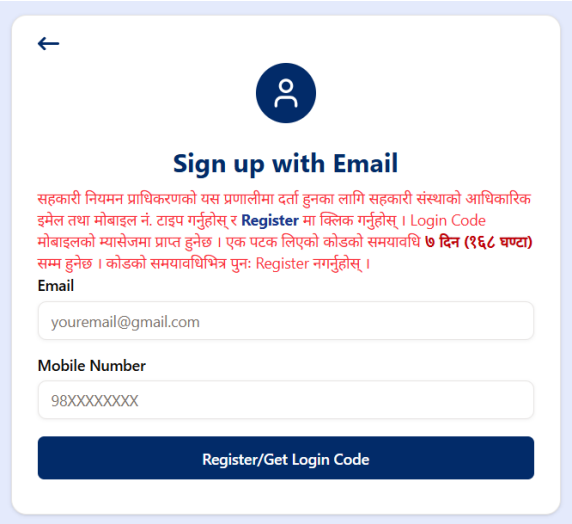


Figure 4.8: Registration

- Integrated authentication APIs for user login and registration.
- Managed user session and token storage using localStorage/sessionStorage.
- Configured protected routes for authenticated users.
- Implemented role-based navigation (Cooperative Admin, Regulator, Super Admin).
- Developed dynamic sidebar menu based on user roles.

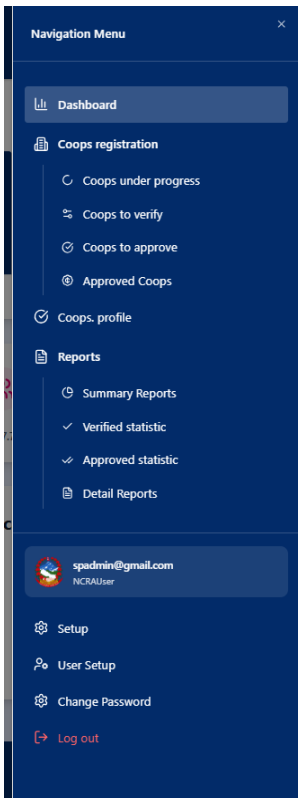


Figure 4.9: Sidebar (Admin)

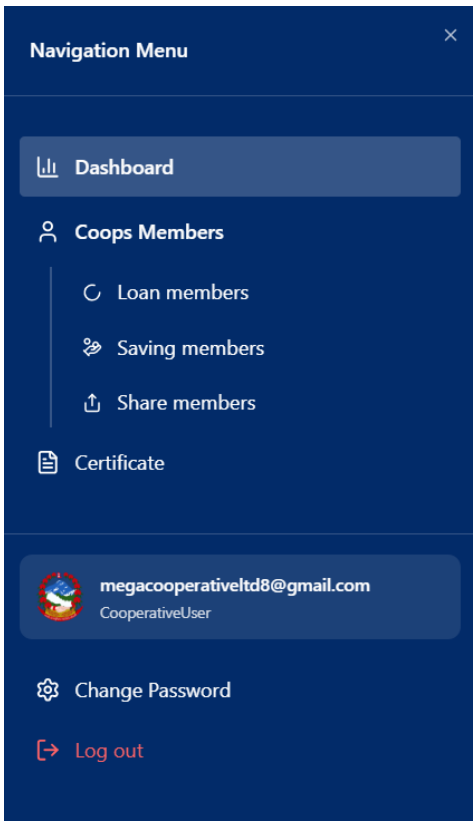


Figure 4.10: Sidebar (User)

- Improved navigation flow and overall user experience.

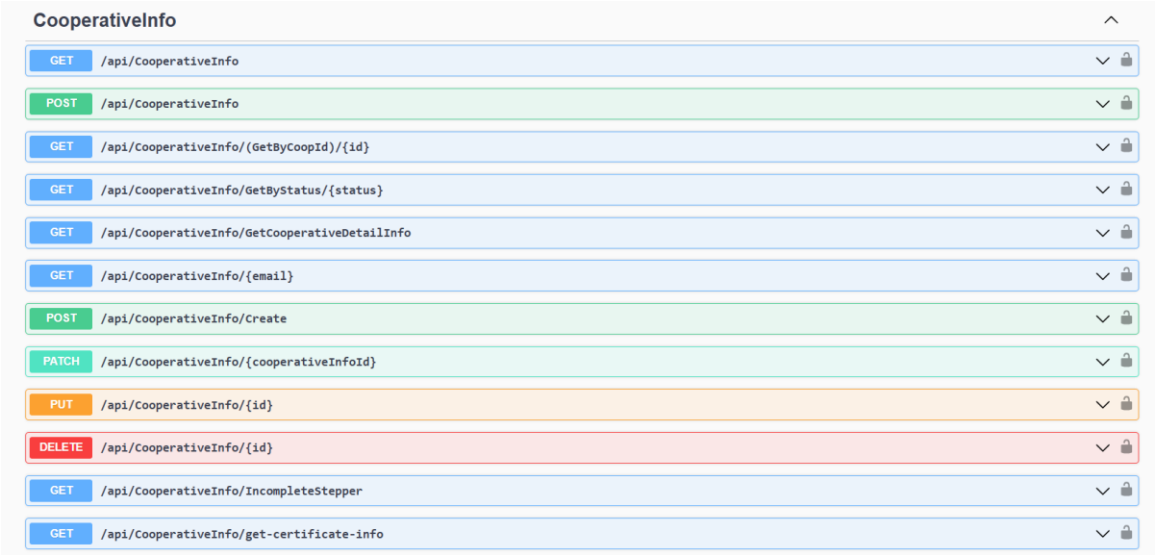


Figure 4.11: Swagger API

iv. Dashboard and Cooperative Registration:

- Built admin dashboard UI with summary cards and statistics.

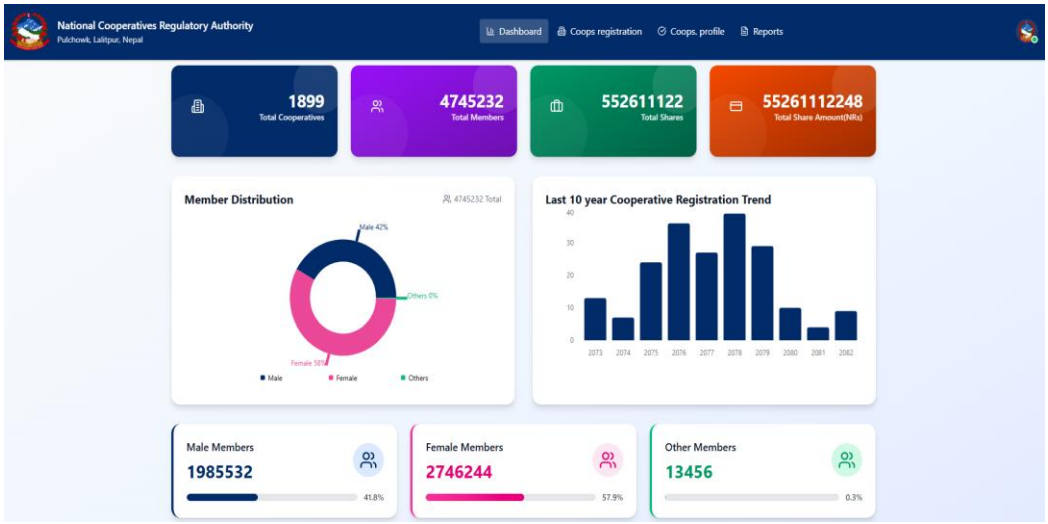


Figure 4.12: Dashboard

- Integrated API for dashboard data display.
- Designed charts and visual components using MUI X Charts.
- Developed cooperative registration form with multi-step stepper using MUI.

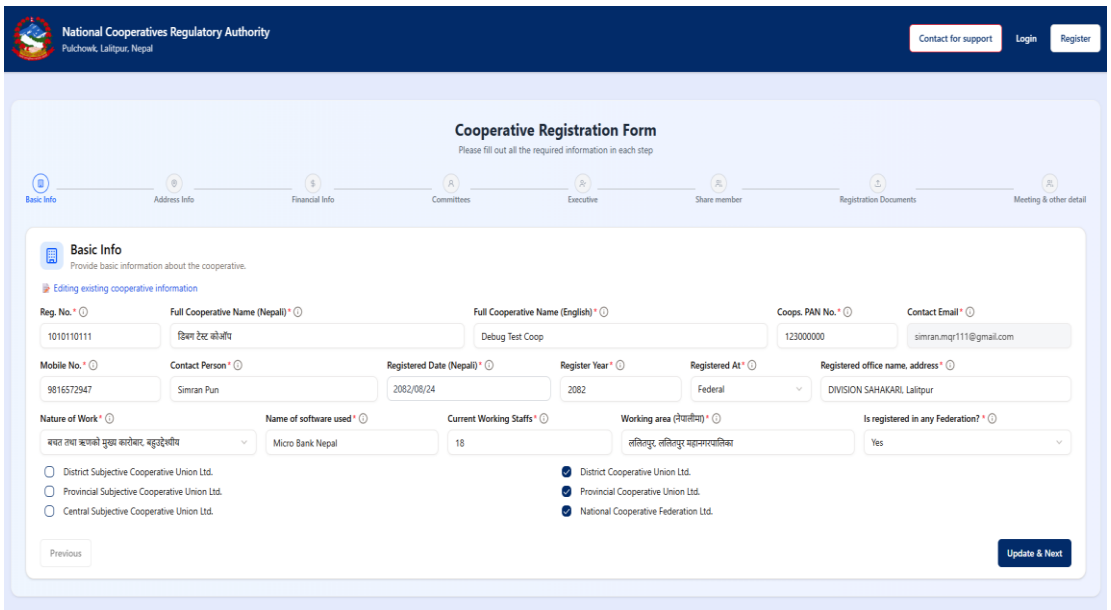


Figure 4.13: Stepper Form

- Managed complex form state using React hooks (useState, useEffect).
- Ensured proper validation across all steps.
- Connected registration forms with backend APIs for data submission.
- Handled API responses, error states, and loading indicators.

v. Performance Optimization and File Upload:

- Implemented reusable hooks and utilities for common functionality.
- Optimized component rendering using React.memo and useCallback.
- Improved TypeScript typings for better type safety.
- Refactored code for better maintainability and readability.
- Implemented file upload functionality with SSL heartbeat for secure transmission.

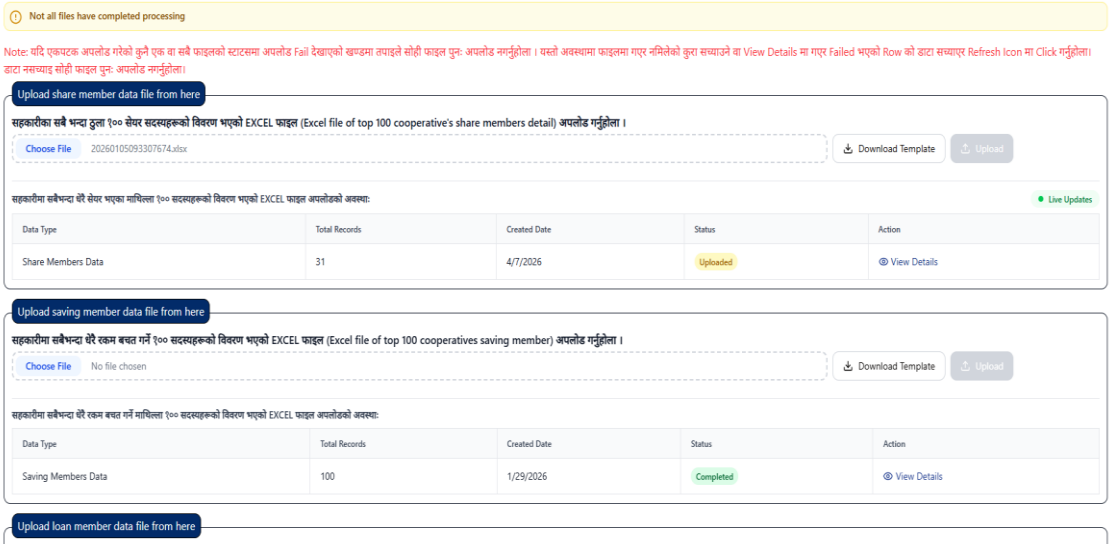


Figure 4.14: File Upload with real-time upload status

- Added preview and validation for uploaded documents.
- Integrated file APIs with real-time upload status indicators.
- Ensured proper error handling for failed uploads.

vi. Testing and Debugging:

- Tested application across different devices (desktop, tablet, mobile).

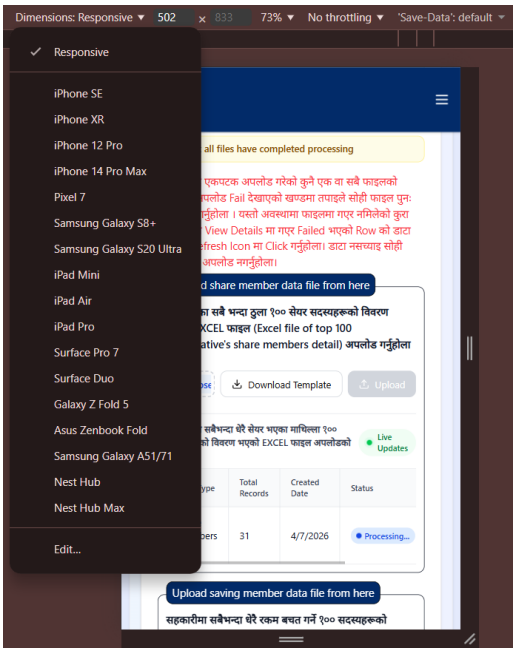


Figure 4.15: Tested application across different devices

- Fixed UI and functional bugs identified during testing.
- Improved responsiveness for semi-urban and rural access.
- Debugged API integration issues and resolved inconsistencies.
- Implemented lazy loading for components to improve initial load time.
- Optimized API calls with caching and debouncing.
- Enhanced UI/UX consistency across all modules.

vii. Admin Verification and Password Management:

- Developed admin verification module with verify/un verify API integration and real-time feedback.

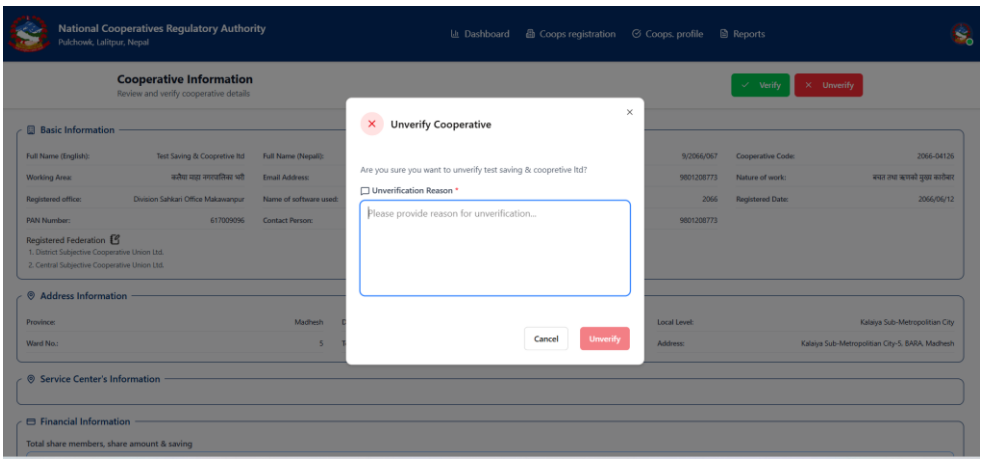


Figure 4.16: Reason for unverification

- Created reusable components to fetch all cooperative data for admin approval, reducing code duplication.
- Built dynamic tables for different cooperative statuses:
  - Registered Cooperative List
  - Verified Cooperative List

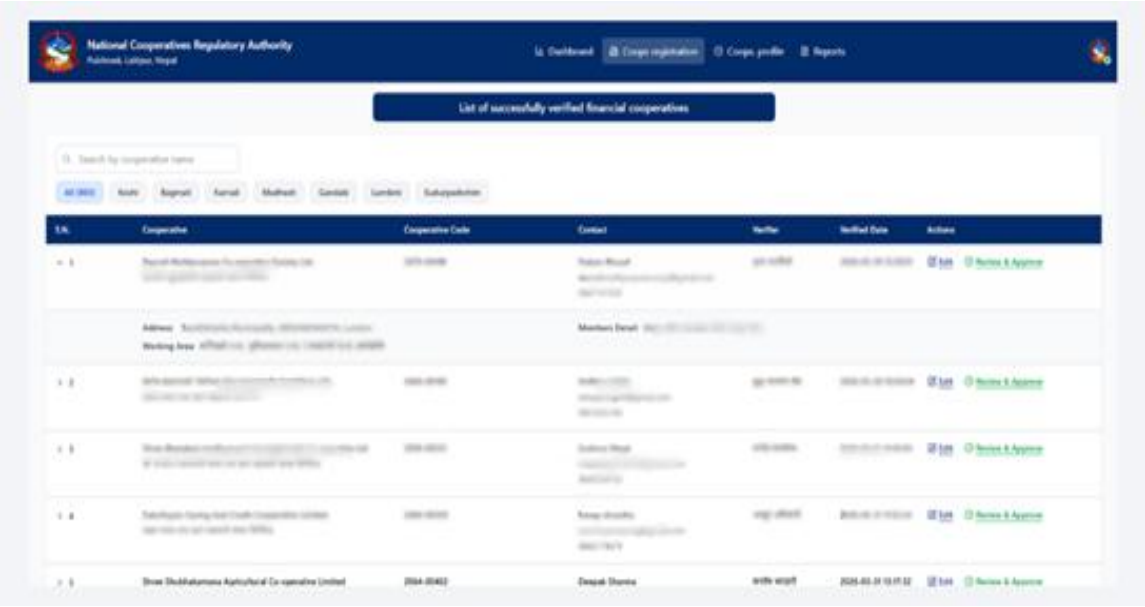
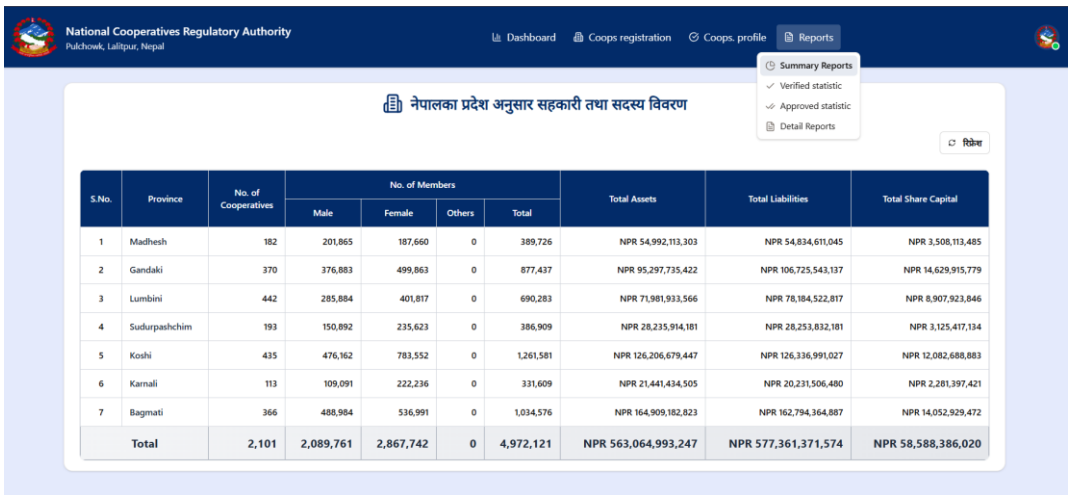


Figure 4.17: Dynamic Table for Cooperative List

  - Approved Cooperative List
  - Reports Table
  - Summary Reports



S.No.	Province	No. of Cooperatives	No. of Members				Total Assets	Total Liabilities	Total Share Capital
			Male	Female	Others	Total			
1	Madhesh	182	201,865	187,660	0	389,726	NPR 54,992,113,303	NPR 54,834,611,045	NPR 3,508,113,485
2	Gandaki	370	376,883	499,863	0	877,437	NPR 95,297,735,422	NPR 106,725,543,137	NPR 14,629,915,779
3	Lumbini	442	285,884	401,817	0	690,283	NPR 71,981,933,566	NPR 78,184,522,817	NPR 8,907,923,846
4	Sudurpashchim	193	150,892	235,623	0	386,909	NPR 28,235,914,181	NPR 28,253,832,181	NPR 3,125,417,134
5	Koshi	435	476,162	783,552	0	1,261,581	NPR 126,206,679,447	NPR 126,336,991,027	NPR 12,082,688,883
6	Karnali	113	109,091	222,236	0	331,609	NPR 21,441,434,505	NPR 20,231,506,480	NPR 2,281,397,421
7	Bagmati	366	488,984	536,991	0	1,034,576	NPR 164,909,182,823	NPR 162,794,364,887	NPR 14,052,929,472
	Total	2,101	2,089,761	2,867,742	0	4,972,121	NPR 563,064,993,247	NPR 577,361,371,574	NPR 58,588,386,020

Figure 4.18: Reports

- Implemented Forgot Password and Reset Password with full API integration and email/OTP verification.

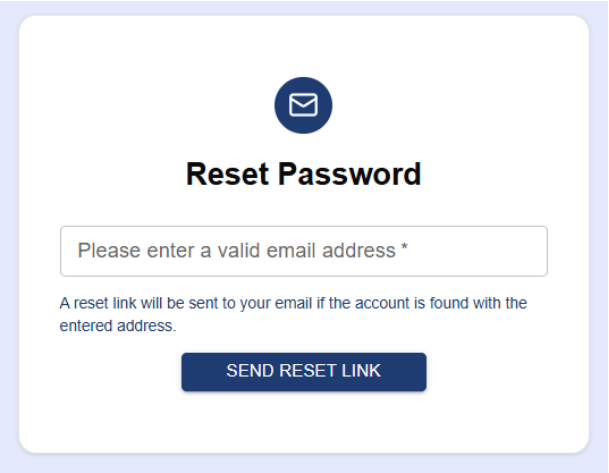


Figure 4.19: Reset Password

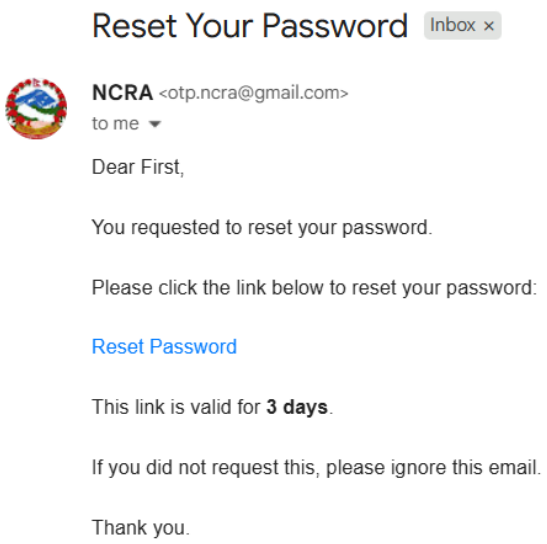


Figure 4.20: Reset Password Link

- Implemented Change Password functionality for authenticated users with API integration.

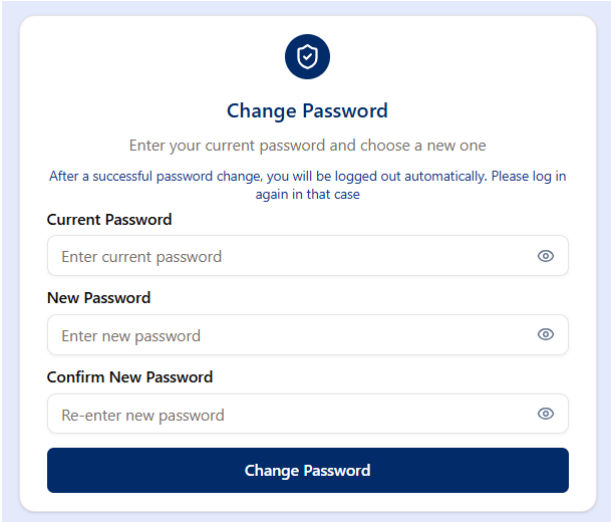


Figure 4.21: Change Password

viii. Form Design with Material UI:

- Installed and integrated the Material UI component library into the FCRS-NCRA project with NCRA branding theme.

```
PS C:\Users\simra\OneDrive\Desktop\FCRS-NCRA> npm install @mui/icons-material @mui/material @emotion/styled @emotion/react
>>
```

Figure 4.22: MUI installation:

- Designed and implemented forms for core FCRS-NCRA modules: Cooperative Registration, Financial Reporting, Compliance Monitoring.
- Developed detailed forms for Cooperative Management components:
  - Financial Management (cooperative basic details, committee/executive member info)

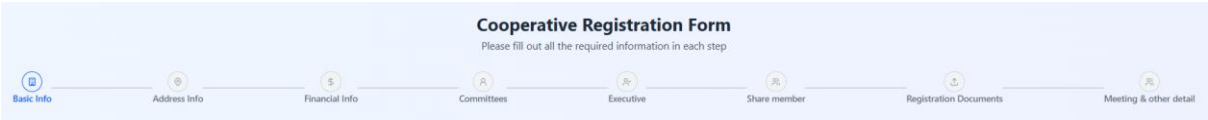


Figure 4.23: Multi-step stepper

  - Meeting details tracking and management
  - Share members, loan members, saving members tracking across fiscal years

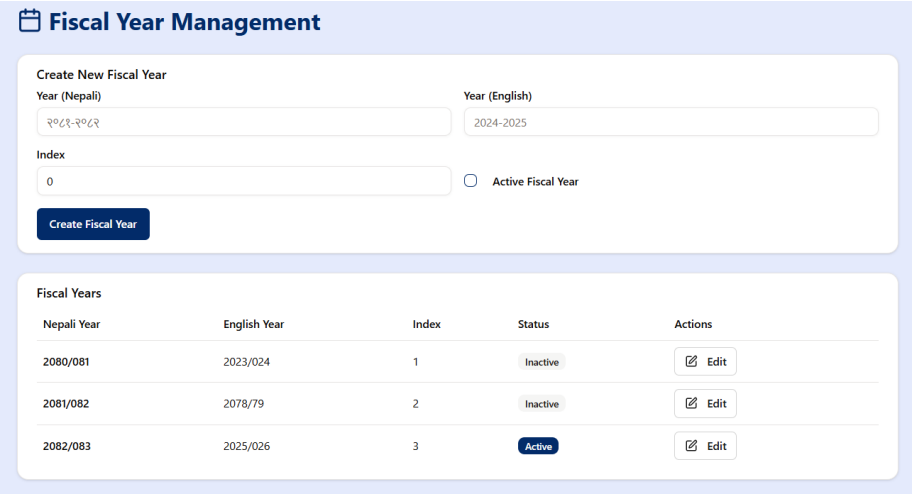


Figure 4.24: Fiscal Year Management

- Complete order tracking and financial transaction history

- Implemented Document Management workflow forms:
  - Document uploads with admin review interface
  - Admin verification/approval workflow (verify/approve/unverify/unapprove with feedback)
  - OTP verification for user authentication

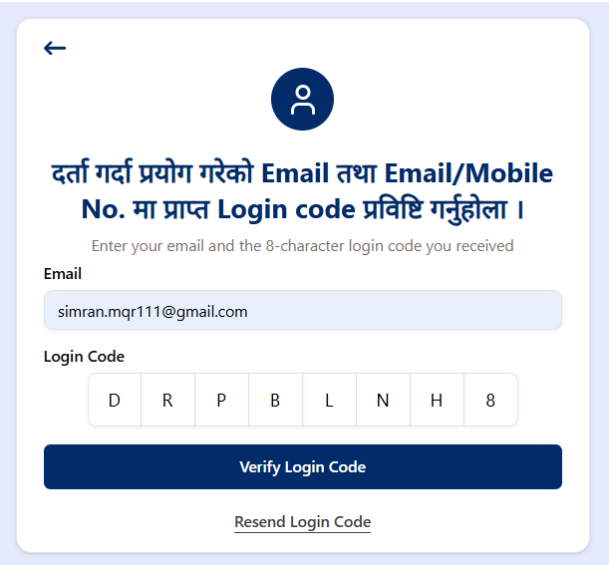


Figure 4.25: OTP Verification for cooperative

  - Certificate generation with cooperative details and signature fetching

Figure 4.26: Dynamic Certificate

- Adapted Material UI designs to match government UI standards.

ix. API Integration – Data Submission and Fetching:

- Implemented Axios for cooperative registration APIs (POST/PUT/GET for basic details, committee members).

Figure 4.27: Swagger API for committee

- Developed OTP verification API integration for certificate generation and document approval.
- Created dynamic dropdown APIs for fiscal years, cooperative categories, member types.
- Implemented real-time status tracking APIs for approval workflows and certificate generation.
- Integrated Forgot Password, Reset Password, and Change Password APIs.
- Used SSL connection for real-time file uploading with heartbeat monitoring.

Figure 4.28: SSL connection

x. Dynamic Table Design

- Built Material UI DataGrid for Cooperative Management tables displaying:
  - Cooperative lists (Registered, To Verify, To Approve, Approved) with all details

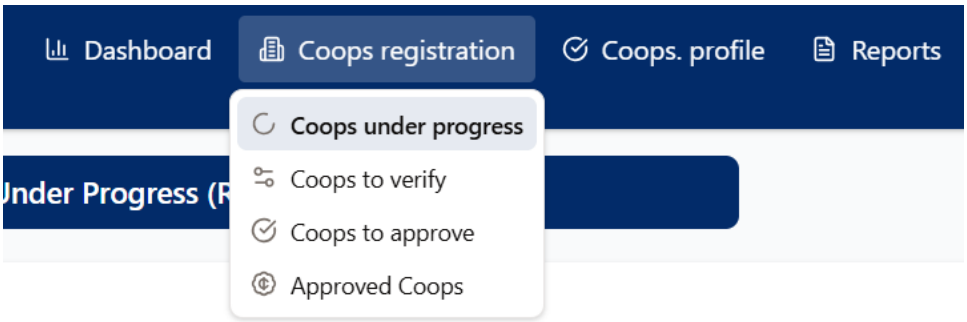


Figure 4.29: Cooperative Table

  - Share/loan/saving member tracking across fiscal years
  - Document approval status with admin feedback
- Added filtering by province, cooperative type, and approval status.

Figure 4.30: Filter By Province

- Implemented actions for certificate generation and status updates.

xi. Report Creation

- Designed Cooperative Performance Reports showing:
  - Financial health metrics (savings, loans, shares by fiscal year)
  - Member growth trends and committee composition
  - Compliance status and document verification rates
  - Regional cooperative performance comparisons

- Created printable PDF reports with NCRA letterhead and digital signatures.

```
src > components > multi-step-form > form-components > review-components > CoopPdf.jsx > useCoopPdfGenerator
39  export const useCoopPdfGenerator = () => {
45    const generatePDF = async (cooperativeData) => {
770       // Save the PDF
771       const fileName = `Cooperative_Review_${coopInfo?.registrationNo || "unknown"}_${new Date().toISOString().split("T")[0]}.pdf`;
772       doc.save(fileName);
773
774       toast.success("PDF generated successfully!");
775     } catch (error) {
776       console.error("Error generating PDF:", error);
777       toast.error("Failed to generate PDF. Please try again.");
778     } finally {
779       setIsGeneratingPDF(false);
780     }
781   };
782
783   return {
784     generatePDF,
785     isGeneratingPDF,
786   };
787 };
788
```

Figure 4.31: Code base for printable PDF reports

- Developed Summary Reports Dashboard consolidating key metrics.

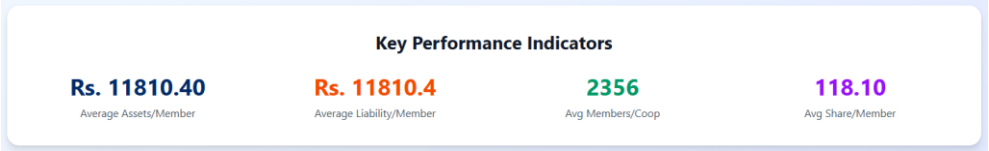


Figure 4.32: Summary key metrics

- Organized data logically into subsections for enhanced readability.

xii. Quality Assurance & Accreditation (QAA) Module Development

- Developed forms for QAA application submission and renewal tracking.
- Implemented document upload and verification workflow for accreditation.
- Created compliance checklist and scoring system.

xiii. Finance Module Refinement and Bug Fixing

- Enhanced Financial Transaction forms for share, loan, savings deposits.

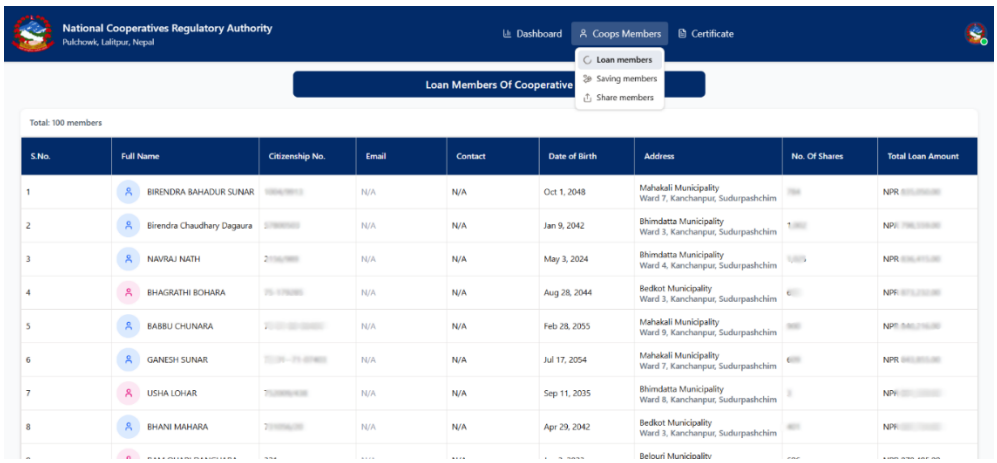


Figure 4.33: Dynamic Tables of member i.e. Loan, Saving, Share

- Implemented fiscal year-wise transaction tracking and reporting.
- Added admin approval workflow for high-value financial transactions.

```
export const useApproveCooperative = () => {
  const queryClient = useQueryClient();

  return useMutation({
    mutationFn: approveCooperative,
    onSuccess: (data) => {
      // Invalidate and refetch relevant queries
      queryClient.invalidateQueries({ queryKey: ['cooperatives'] });
      queryClient.invalidateQueries({ queryKey: ['cooperative-detail'] });
      queryClient.invalidateQueries({ queryKey: ['cooperatives-to-approve'] });
    },
    onError: (error) => {
      console.error('Failed to approve cooperative:', error);
    },
  });
};
```

Figure 4.34: Code base to approve cooperative

- Fixed responsive issues in financial management forms across devices.
- Optimized API performance for large member datasets.
- Resolved certificate signature fetching issues.

```
useEffect(() => {
  const fetchSign = async () => {
    try {
      const response = await axios.get(
        `${BASE_ENDPOINT}/admin/CooperativeApproval/by-coop?coopId=${coopId}`,
        {
          headers: {
            Authorization: `Bearer ${token}`,
          },
        }
      );
      const approverId = response.data?.[0]?.approvedByUserId;
      setUserSignId(approverId || null);
    } catch (err) {
      console.error("Failed to fetch sign info:", err);
    }
  };
  fetchSign();
}, [coopId, token]);
```

Figure 4.35: Code base to fetch signature

- Enhanced document approval workflow with better feedback mechanisms.

xiv. Final Deployment

- Performed final code cleanup and refactoring.
- Reviewed all modules and features for completeness.
- Assisted in deployment setup and production release.

These contributions not only improved The Author's technical skills but also added value to the FCRS-NCRA project by making it more user-friendly, efficient, and effective in addressing its objectives.

Code base:

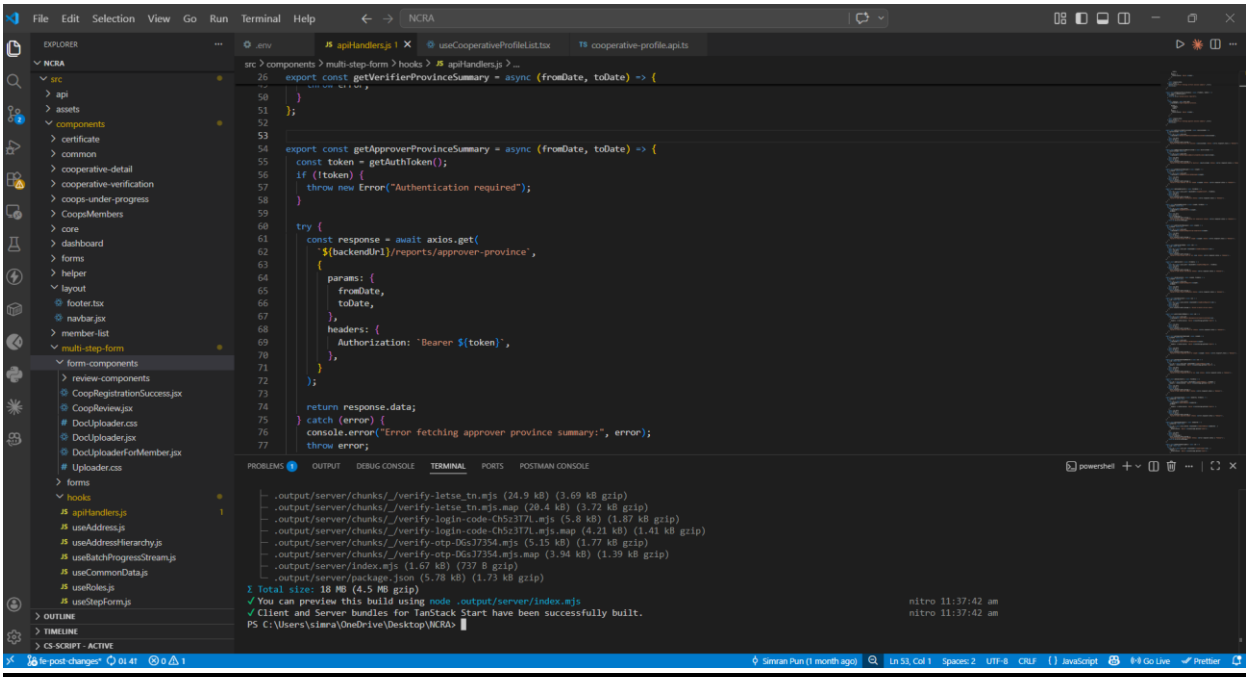


Figure 4.36: Code base of the project

CHAPTER 5: CONCLUSION AND LEARNING OUTCOME

5.1 Conclusion

The FCRS-NCRA frontend system delivers a robust, responsive web application that streamlines cooperative registration, financial management, and regulatory compliance across Nepal's cooperative sector. Users can efficiently manage cooperative infrastructure, generate certificates with digital signatures, track member financial data (shares, loans, savings), and process document approvals through intuitive interfaces accessible at https://fcrs.ncra.gov.np. This platform, developed for the National Cooperative Regulatory Authority, features modern React.js interfaces with Material UI that support seamless interaction for administrators, regulators, and cooperative staff nationwide (N. C. R. Authority, 2074) .

The frontend system underwent rigorous testing across devices and has been successfully deployed, meeting NCRA's requirements for government-grade usability and performance. This internship significantly enhanced my frontend development expertise while providing deep insights into cooperative governance technology needs. It underscored effective communication for API coordination with backend teams, highlighted collaborative development practices, and demonstrated transforming complex regulatory requirements into practical, user-centered software solutions.

5.2 Learning Outcomes

i. Proficiency in Frontend Technologies

Gained hands-on mastery of React.js, Material UI, and Axios through FCRS-NCRA development. Built component-based architectures for cooperative management modules. Designed responsive forms and DataGrids for financial ledgers, member tracking, and document workflows. Integrated secure APIs for real-time data exchange, deepening understanding of modern frontend ecosystems and government UI standards.

ii. Advanced UI Features Development

Mastered implementation of sophisticated features enhancing FCRS-NCRA workflows:

- Print functionality using @react-to-print for registration certificates and financial reports.
- Dynamic certificate templates auto-populating cooperative/member financial data with digital signatures.
- Drag-and-drop document uploads with admin approval workflows and OTP verification.
- These innovations improved regulatory efficiency and user experience for NCRA stakeholders.

iii. Collaboration and Teamwork

Collaborated with backend developers through Agile sprints, learning to:

- Align frontend components with API contracts via Swagger documentation.
- Conduct code reviews maintaining Material UI consistency and react best practices.
- Solve cross-functional challenges like responsive design conflicts and performance optimization.
- Teamwork proved essential for delivering production-ready government software.

iv. Time Management and Multitasking

Managed parallel development across FCRS-NCRA modules (financial management, QAA, certificates) by:

- Prioritizing tasks based on NCRA deployment timelines and regulatory priorities.
- Breaking complex workflows into weekly deliverables (forms → APIs → tables → reports).
- Maintaining consistent progress through daily stand-ups and sprint reviews.
- This ensured all frontend responsibilities met deadlines without compromising quality.

v. Understanding Real-World Project Workflows

Gained exposure to professional software delivery:

- Agile practices including sprint planning, retrospectives, and NCRA stakeholder demos.
- Git workflows for feature branches, pull requests, and production deployments.
- Complete SDLC from wireframes → development → testing → government approval.
- These prepared me for enterprise frontend roles in public sector projects.

vi. Enhanced Communication Skills

Improved professional interactions by:

- Explaining React component architecture and API integration to non-technical NCRA stakeholders.
- Documenting FCRS-NCRA modules for team knowledge sharing and maintenance.
- Providing constructive feedback during peer reviews of UI/UX implementations.
- These skills strengthened collaborative delivery in regulated government environments.

REFERENCES

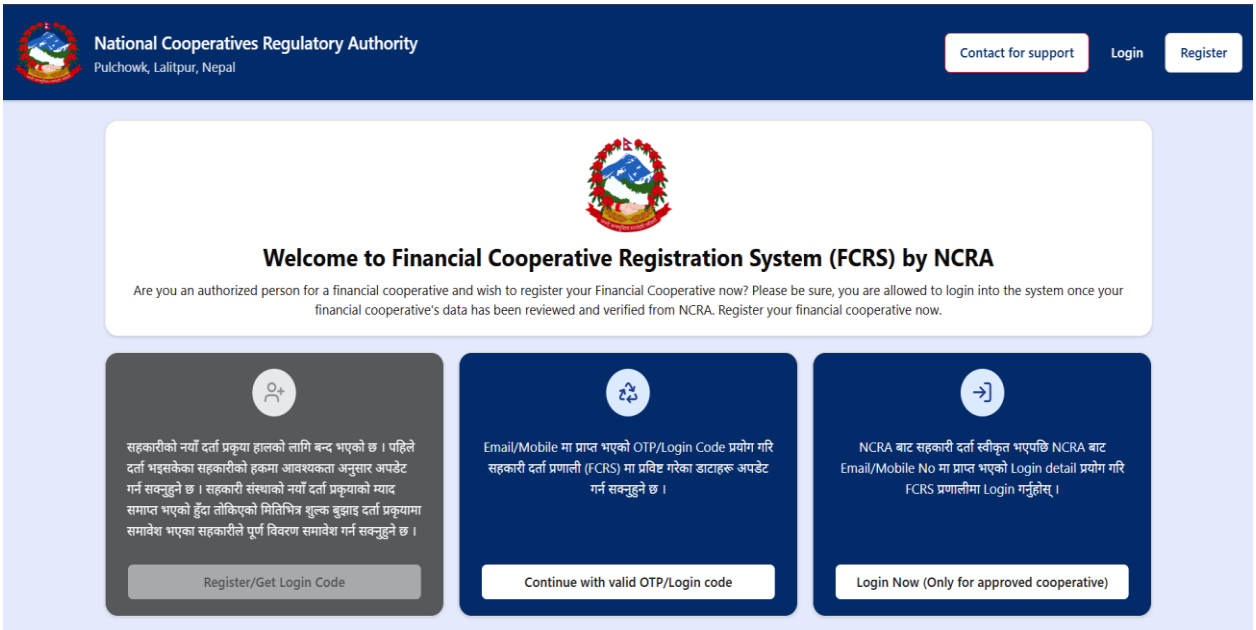
Authority, N. C. R. (2074). Cooperatives. https://fcrs.ncra.gov.np/

Ltd., D. P. (2082). National Cooperative Regulatory Authority, Financial Cooperative Registration System (FCRS) user manual. https://ncra.gov.np/content/13343/cooperative-user-manual/

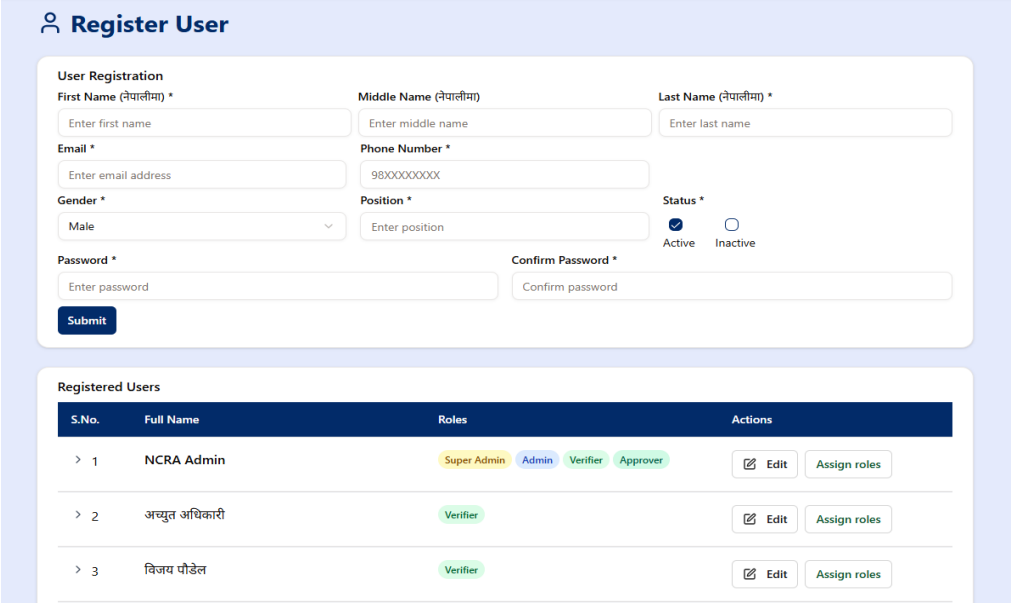
National Cooperative Regulatory Authority. (2025, November 27). Rising Nepal Daily. https://risingnepaldaily.com/news/71754

APPENDIX

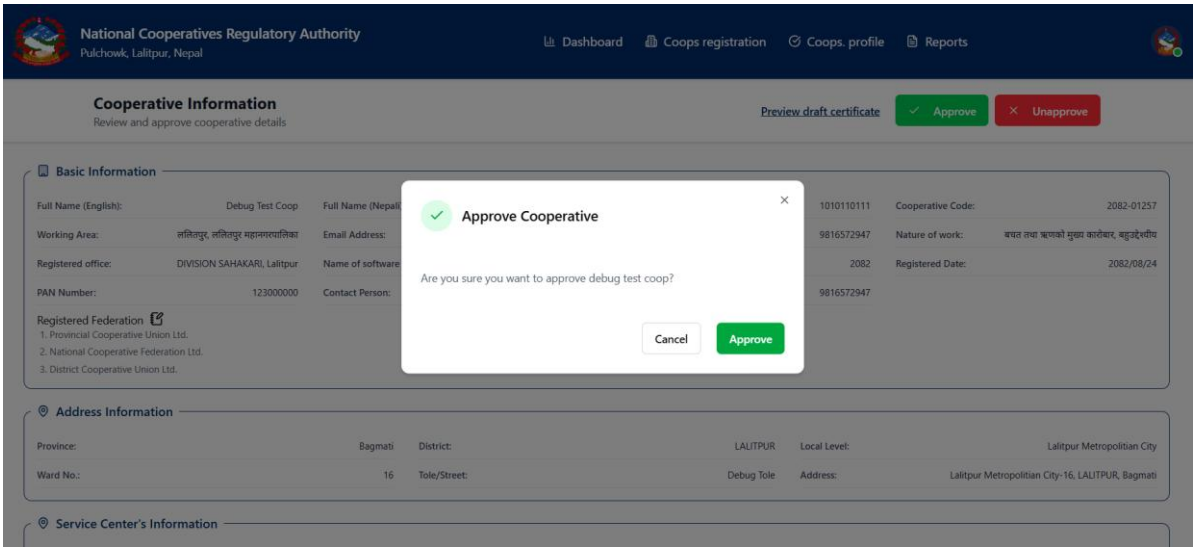
Landing Page:



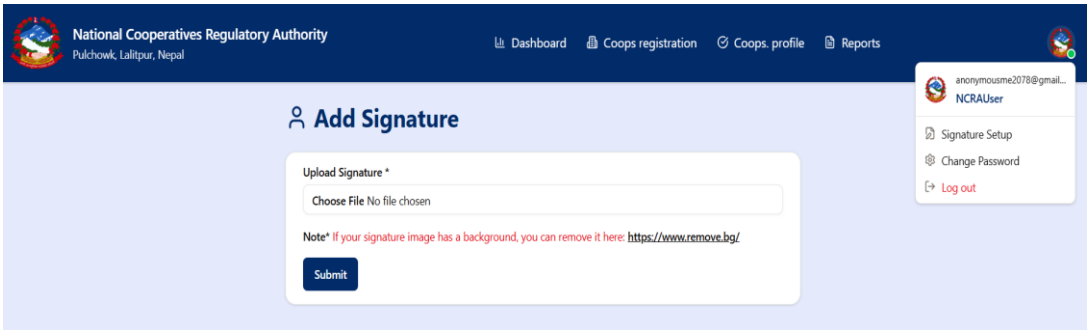
Add User-based role:



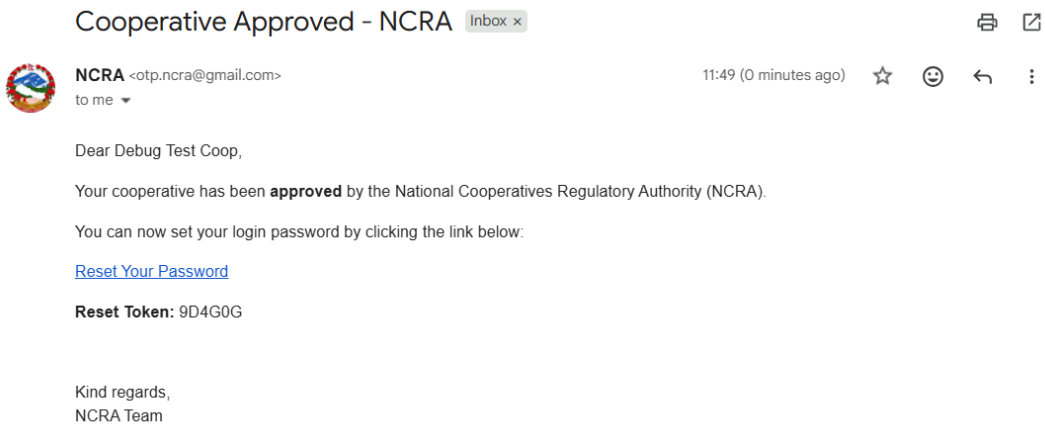
Admin Approval Portal:



Add Signature:



Approved mail:



Cooperative Approved - NCRA Inbox ×

NCRA <otp.ncra@gmail.com> 11:49 (0 minutes ago)
to me

Dear Debug Test Coop,

Your cooperative has been **approved** by the National Cooperatives Regulatory Authority (NCRA).

You can now set your login password by clicking the link below:

Reset Your Password

Reset Token: 9D4G0G

Kind regards,
NCRA Team